

Fire detector alarm

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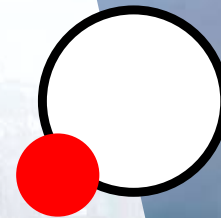


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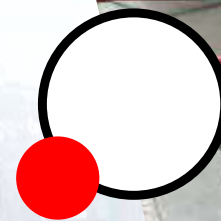
Scope:

- Detects fire by sensing temperature rise using a thermistor. Converts temperature changes into an electrical signal.
- Activates a buzzer alarm when the temperature exceeds a certain level.
- Can be used in homes, schools, offices, shops, and labs.
- Useful as an early fire-warning system.
- Good for students learning electronics, especially sensors, transistors, and 555-timer IC operation.
- Can be expanded with LED indicators, relays, or GSM modules.



Objectives:

- To detect a temperature rise that indicates a possible fire.
- To convert temperature changes into a voltage signal.
- To trigger the 555 Timer IC when temperature crosses a threshold.
- To activate a buzzer alarm to warn people.
- To provide a simple, low-cost, and reliable safety device.
- To understand the working of sensors, potentiometer, transistor, and 555 IC.



Components used:

1. Power Source

9V Battery

2. Sensors

NTC Thermistor (10k Ω at 27°C)
temperature sensor

3. Resistors

R1 – 100k Ω

R4 – 4.7k Ω

R5 – 1k Ω

R6 – 100k Ω

4. Variable Resistor

Potentiometer – 1M Ω (50%)
used for sensitivity adjustment

5. Semiconductors

NPN Transistor (100 A/A gain)

6. IC

555 Timer IC

7. Capacitors

C1 – 10 μ F electrolytic capacitor

8. Output Device

Buzzer (200 Hz)

9. Wires & Breadboard/PCB

Connecting wires

Breadboard or PCB (optional but needed
for building the circuit)

Function of each component:

1. Thermistor (Temperature Sensor)

A temperature-sensitive resistor whose resistance decreases as the temperature increases, allowing the circuit to detect heat changes.

2. Potentiometer (1M Ω)

Allows manual adjustment of sensitivity by changing the resistance level in the sensor circuit.

3. R1 (100k Ω), R4 (4.7k Ω), R5 (1k Ω) & R6 (100k Ω)

Control and limit electrical current in different parts of the circuit to ensure safe and correct operation.

4. Transistor (NPN)

Acts as a switch or amplifier that turns on the output (like the buzzer) when triggered by the sensor signal.



5. 555 Timer IC

Generates timed pulses or signals and is used here to control when the buzzer activates.

6. Capacitor C1 (10 μ F)

Stores and releases small amounts of energy to stabilize signals or create timing delays in the circuit.

7. Buzzer

Produces an audible sound when activated, alerting the user.

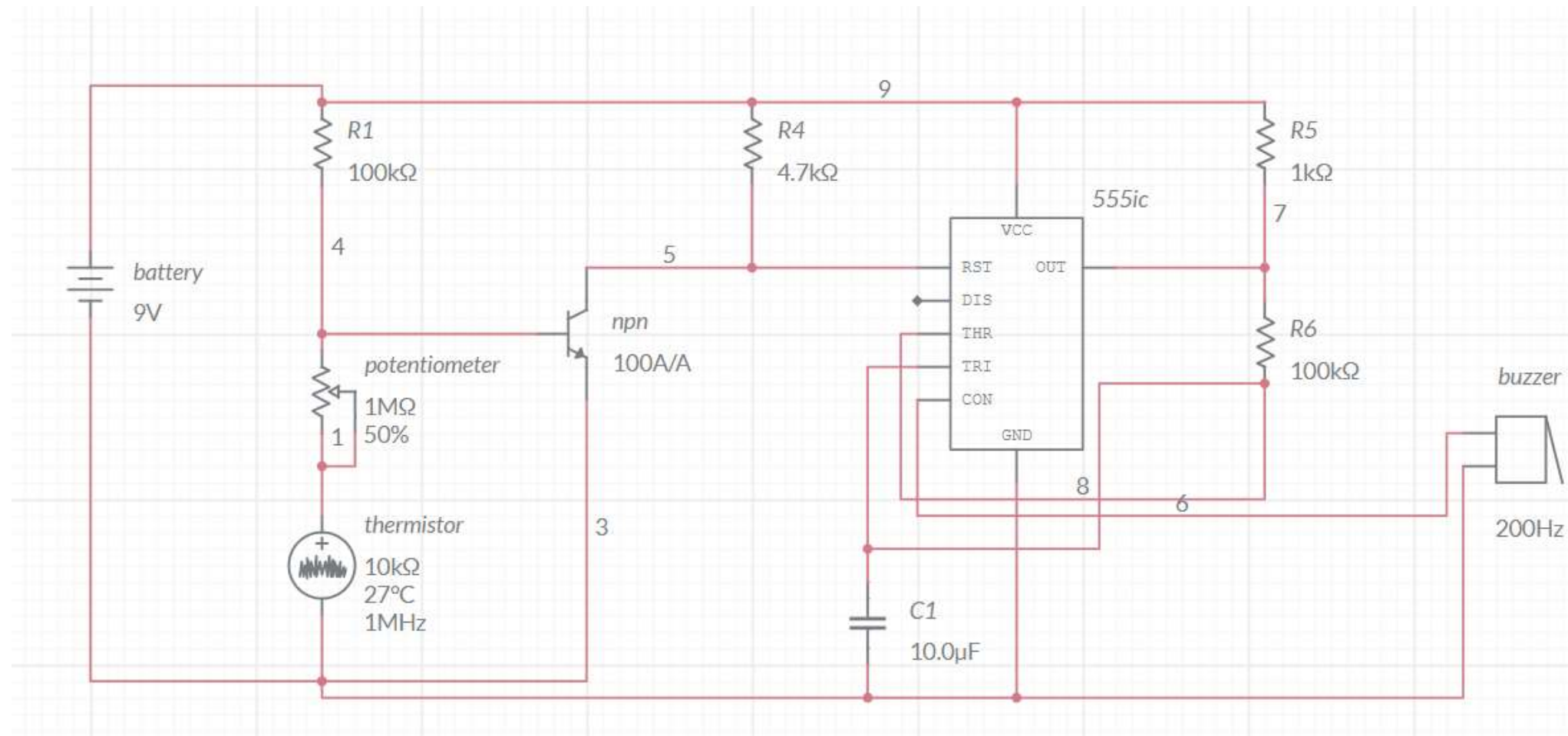
8. 9V Battery

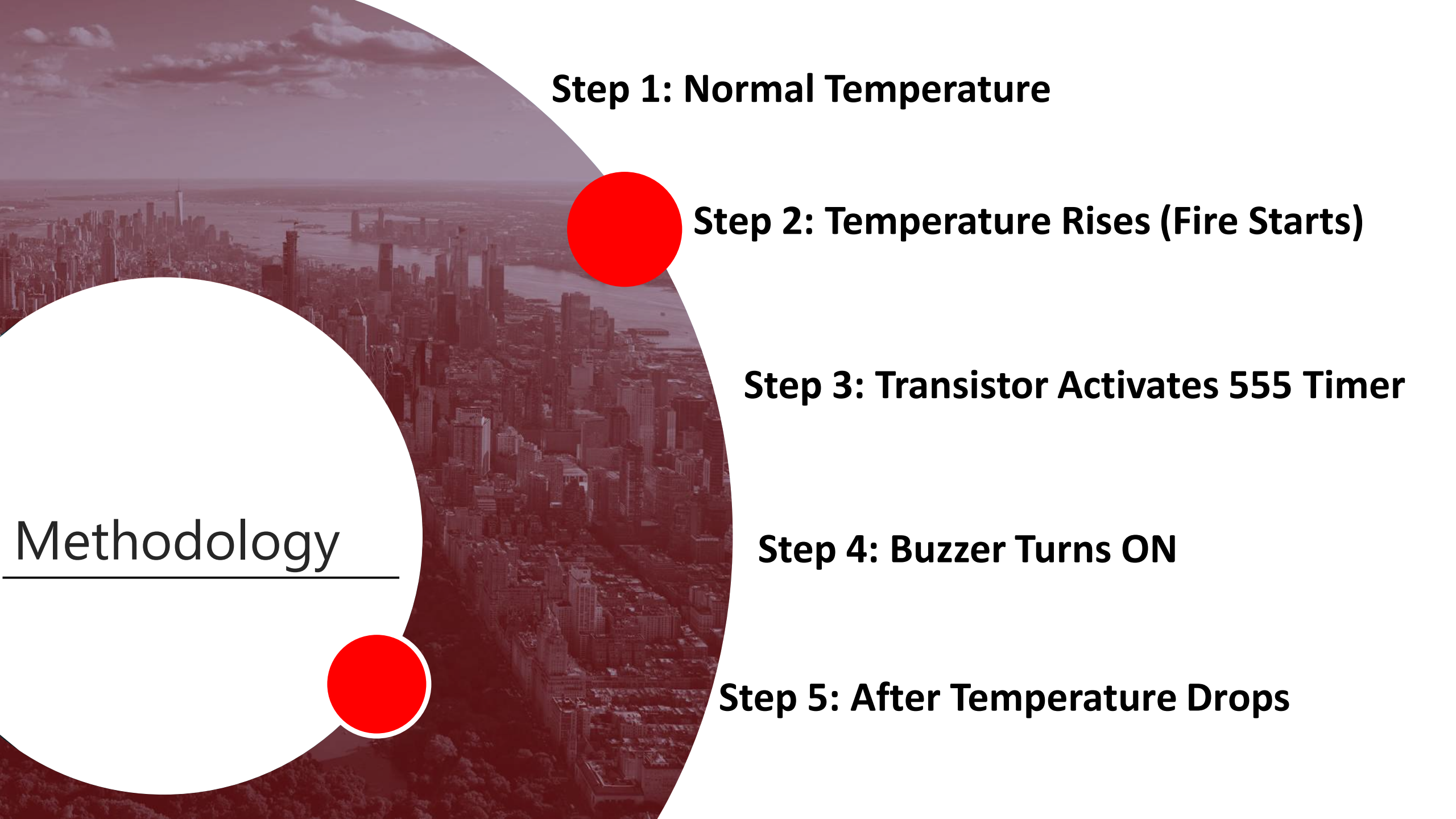
Provides the electrical energy needed to power the entire circuit.



Circuit Diagram:

Circuit Multisim Link: <https://www.multisim.com/content/EFACdFiqjDuF9CUUPupaFK/fire-alarm-using-thermistor-1/open/>





Step 1: Normal Temperature



Step 2: Temperature Rises (Fire Starts)

Step 3: Transistor Activates 555 Timer

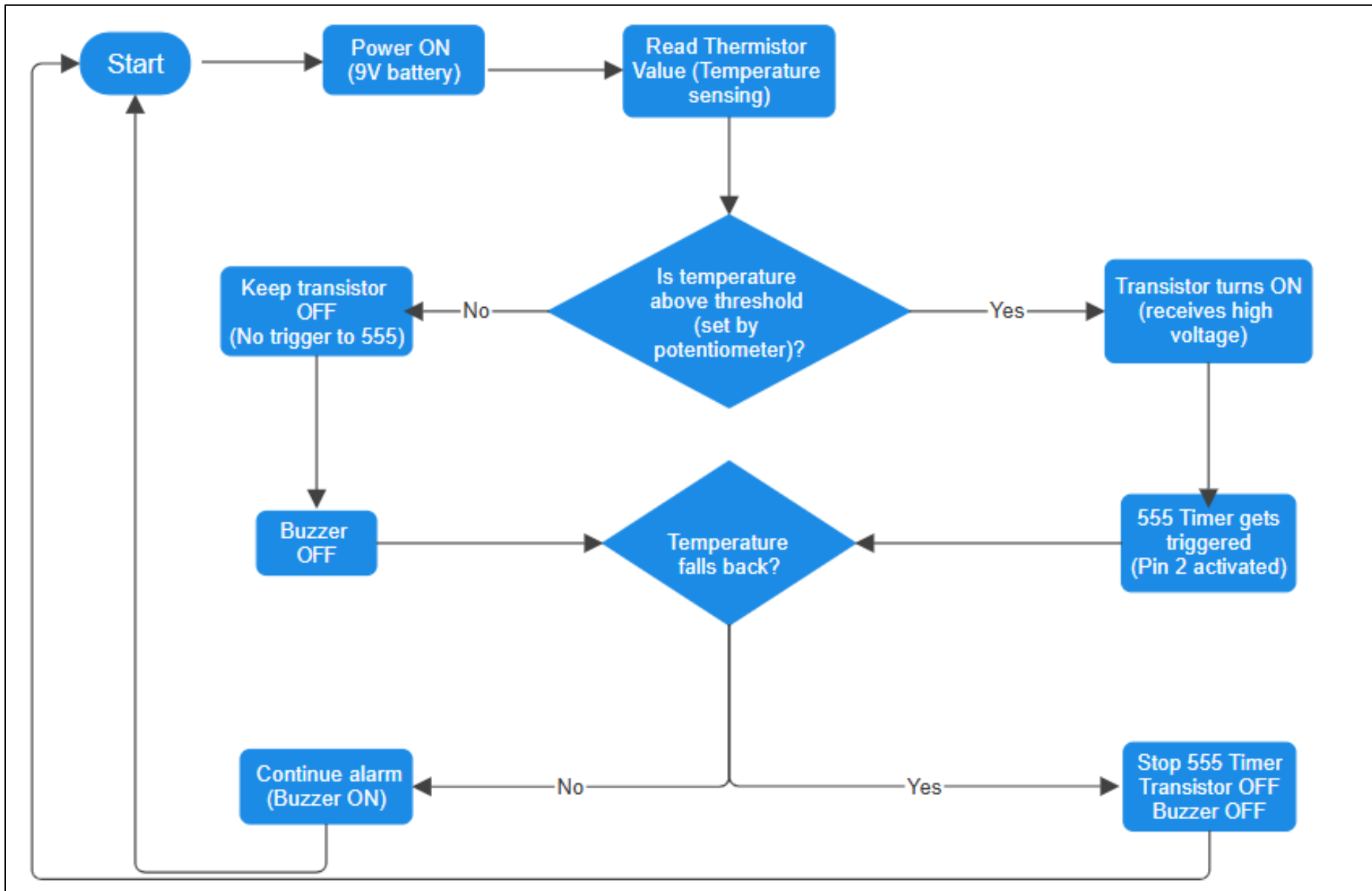
Step 4: Buzzer Turns ON



Step 5: After Temperature Drops

Methodology

Flow chart:



Limitations:

Limited Input Voltage/Current Range

The circuit may only work correctly within a specific voltage or current range.

If the input exceeds the rating of the components (resistors, comparator, microcontroller), it can cause:

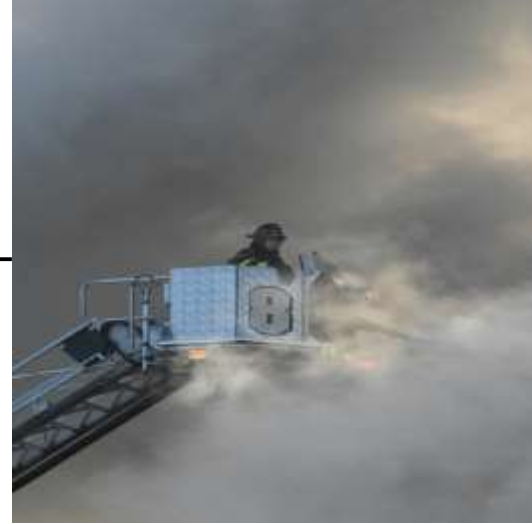
- inaccurate readings
- component heating
- permanent damage

Sensitivity to Noise

Electrical noise from surrounding circuits or the power Supply can affect:

- comparator accuracy
- sensor output
- threshold detection

This can cause the circuit to trigger false "More" or "Less" conditions.





Tolerance Errors in Components

Resistors, sensors, and analog components have tolerances (e.g., $\pm 5\%$).

This means:

calculated values may deviate from actual values

threshold levels may not be precise

Delay in Response

Depending on the components used:

sensor response time

comparator propagation delay

microcontroller sampling delay

The circuit may not respond instantly, especially for fast-changing signals.

Overheating Under High Load

If current is high, some components (like shunt resistors)

may heat up, causing:

drift in measurement

component failure

inaccurate voltage drop readings

No Protection Against Short Circuit or Overcurrent

If the design lacks:

fuses

current limiting resistor

overvoltage protection

the circuit can be easily damaged.

Requires Calibration

If sensors or thresholds are not calibrated:

incorrect decisions ("HIGH" or "LOW")

unstable output

Calibration is needed for accuracy.

Cannot Detect AC Without Extra Components

If the circuit is designed for DC only, it cannot measure

AC signals unless:

rectifier

filtering system

RMS converter

are added.



Ways to improve this circuit:

1. Make it more stable

Add small capacitors and better voltage control so the alarm doesn't turn ON/OFF by mistake.

2. Use a better temperature detector

Replace the thermistor with a more accurate sensor like LM35 or DS18B20.

3. Add a comparator

Use LM393 or an op-amp to compare temperature properly. This gives a cleaner and more accurate trigger.

4. Stop false alarms

Add hysteresis (small feedback) or filtering so the alarm doesn't flicker when the temperature is near the limit.

5. Use a MOSFET/transistor to drive the buzzer

This protects the 555 timer and allows louder alarms.



6. Add safety features

Reverse-polarity protection, fuse, and battery low indicator to prevent damage or weak operation.

7. Add buttons

A "Test" button to check the alarm and a "Mute" button to silence it temporarily.

8. Add a smoke or gas sensor

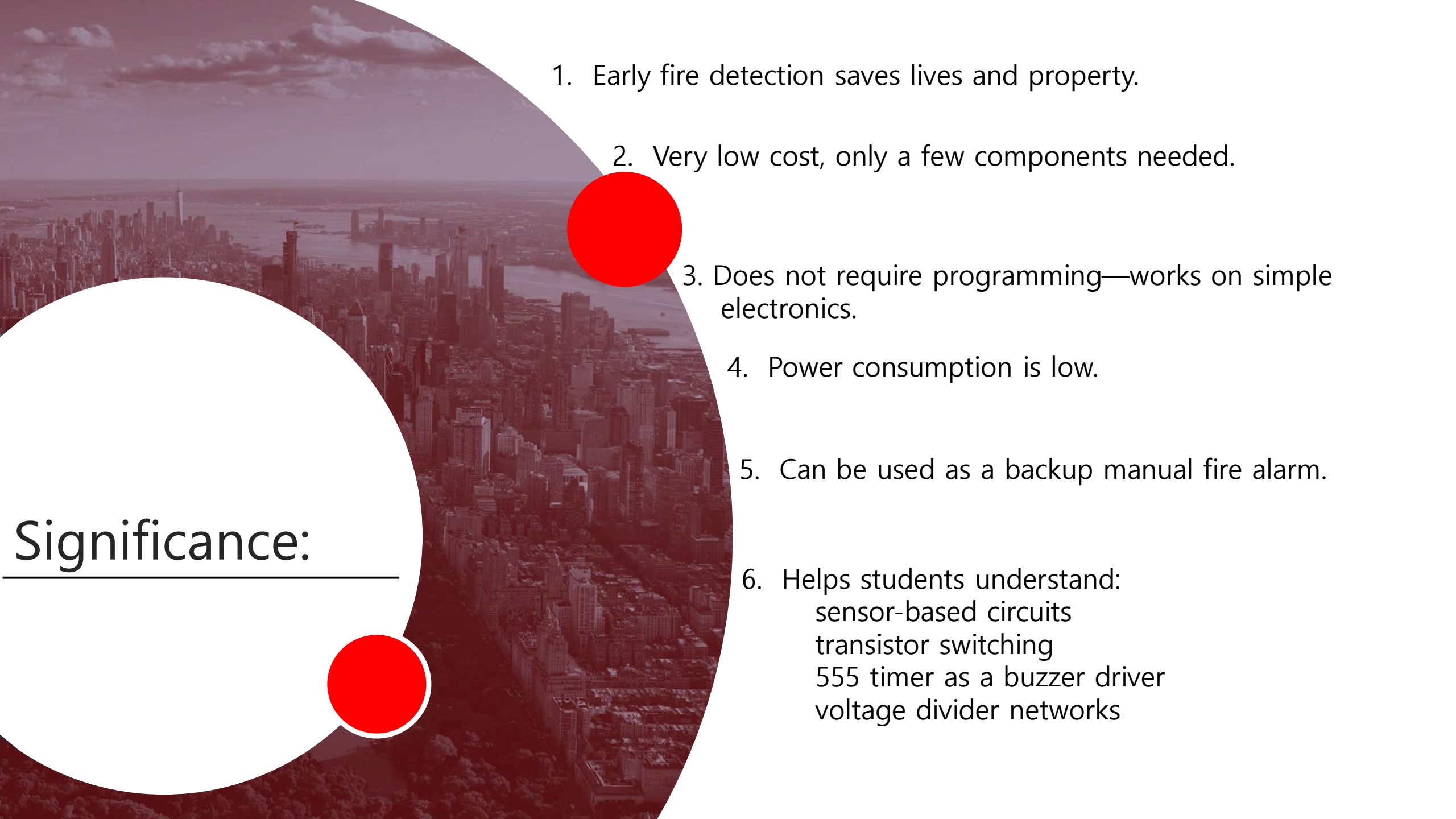
Mix smoke + heat sensing to detect fire more reliably.

9. Use a microcontroller (optional)

A small MCU can read sensors, filter signals, reduce false alarms, and even send alerts.

10. Improve physical design

Use a proper case, keep the sensor exposed to air, and protect the circuit from dust and humidity.

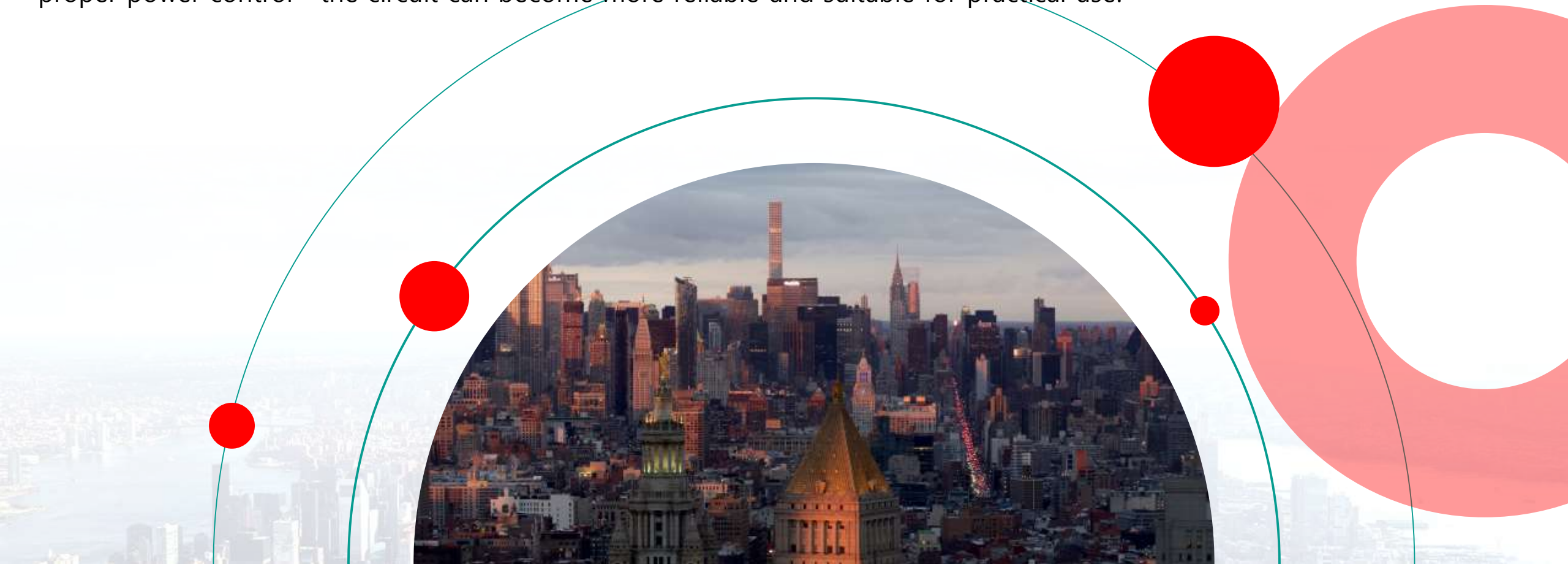


Significance:

1. Early fire detection saves lives and property.
2. Very low cost, only a few components needed.
3. Does not require programming—works on simple electronics.
4. Power consumption is low.
5. Can be used as a backup manual fire alarm.
6. Helps students understand:
 - sensor-based circuits
 - transistor switching
 - 555 timer as a buzzer driver
 - voltage divider networks

Conclusion:

The fire detector alarm circuit is a simple and effective system that uses a thermistor to sense rising temperature and a 555 timer to activate a buzzer when a fire is detected. It provides an affordable and easy-to-build method for basic fire safety. Although it works well for small applications, its performance can be affected by noise, battery weakness, and limited sensing accuracy. With small improvements—such as better sensors, noise filtering, and proper power control—the circuit can become more reliable and suitable for practical use.





THANK YOU

ANY QUESTIONS?

References:

Websites and tools used:

- Canva slides
- Grammarly
- Snappify
- AI GPT
- NI Multisim
- Flowchart maker
- PPT mon
- Meta Lama
- Quill Bot
- Gamma AI
- YouTube
- Physics circuits core syllabi and contents



Timeline

